Capstone Three — Project Proposal

Optimizing Term-Deposit Marketing Campaigns for a European Bank

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Problem statement

A Portuguese retail bank ran a phone-based direct marketing campaign to convert existing customers into term-deposit holders and observed a baseline conversion rate of roughly 11 percent — meaning about nine of every ten calls produced no revenue but still consumed call-center capacity. Using the campaign's historical outcomes together with client and macroeconomic attributes, can we build a targeting model that predicts, before a call is placed, which customers are most likely to subscribe — so that the next campaign contacts fewer, higher-probability customers and lifts realized ROI over a call-everyone baseline?

Context

Direct marketing remains one of the largest recurring cost lines in retail banking. Every outbound call has fixed cost (agent time, telephony, systems overhead), and current targeting relies mostly on customer segmentation rules maintained by hand. When the base conversion rate on a product is as low as 11 percent, even a modest lift in precision at the top of the targeting list translates directly into either fewer calls at the same revenue or the same call volume at meaningfully higher revenue. This problem also mirrors the client-analytics work my team at JPMorganChase does for wealth-management outreach, giving the analysis clear real-world grounding.

Criteria for success

The project succeeds if it produces a reproducible binary classification model that meaningfully outperforms two baselines on held-out data: a majority-class predictor and a simple logistic regression on the top three intuitive features. Performance will be measured with ROC AUC, precision at the top decile of predicted probabilities (the operational metric marketers care about), and expected profit at the model's best threshold given assumed values for cost per call and revenue per subscription. Beyond raw performance, the model must produce interpretable feature attribution using SHAP values, and the written report must include a lift chart and a threshold-versus-profit chart clear enough that a non-technical marketing lead could make a defensible go or no-go decision.

Scope of solution space

Analysis covers all 41,188 client records in the UCI Bank Marketing dataset, capturing 45 months of campaign activity between May 2008 and November 2010. Target variable is the binary subscription outcome. The modeling approach starts with logistic regression as an interpretable baseline, then advances to gradient-boosted tree ensembles (XGBoost and LightGBM) as the primary models, with SHAP for interpretation. Scope explicitly excludes several things: the call duration feature (which perfectly leaks the outcome since a duration of zero implies no subscription); any real-time streaming inference (the deliverable is a batch scoring model); and any personalization to non-Portuguese markets. The project does not attempt to explain why particular macroeconomic conditions drive subscription — only to use them as features when they add signal.

Constraints

Four constraints shape the scope. First, the data is 2008–2010, meaning it spans the acute phase of the 2008 financial crisis but is now 15+ years old; conclusions about specific customer segments should be presented as illustrative of methodology rather than as current market truth. Second, there is a moderate class imbalance (about 11 percent positive), which requires care in evaluation metric choice (accuracy is misleading) and possibly class-weight handling during model fitting. Third, the dataset contains no customer-level financial history beyond a few campaign contact fields — no account balances, no transaction data, no full CRM view — which caps the model's ceiling relative to what a real bank could achieve with its own internal data. Fourth, project time budget is roughly 80 to 120 hours across wrangling, EDA, modeling, and reporting.

Stakeholders

Five stakeholder audiences would use the model or the analysis. Marketing analytics owns the targeting logic and is the primary consumer of the ranked prospect list. Call center operations uses the list to prioritize agent time and to size next-campaign staffing. The term-deposit product team gets qualitative signal about which client attributes correlate with subscription. Finance uses the projected lift for revenue-forecasting and marketing-ROI attribution. Executive leadership reviews the summary charts (lift, threshold-versus-profit) when making channel-mix decisions across direct marketing, digital, and branch. A secondary audience is external — the project produces a portfolio-ready writeup that a hiring manager evaluating a marketing-analytics or customer-analytics role could review.

Data sources

- UCI Bank Marketing dataset — 41,188 client records across 20 input columns plus the binary target. Client attributes include age, occupation, marital status, education, and credit/loan status. Campaign attributes include contact channel, month and day of last contact, days since prior contact, and prior-campaign outcome. Macroeconomic attributes include employment variation rate, consumer price index, consumer confidence index, and Euribor 3-month rate. Primary data source for the project.
- Kaggle mirror (single CSV, easy download) — Kaggle-hosted version of the same file; useful for a direct CSV download without navigating the UCI archive interface.
- Reference paper — Moro, Cortez & Rita (2014) — The original 'Data-Driven Approach to Predict the Success of Bank Telemarketing' publication in Decision Support Systems, cited by the UCI dataset page. Provides the business context and prior modeling baselines to compare against.

Approach

Work follows the standard Data Science Method sequence used in Capstone Two. Wrangling handles categorical encodings, 'unknown' sentinels, and the deliberate exclusion of the leaky call-duration column. EDA covers univariate distributions, subscription rate by feature bin, and correlations against the target. Preprocessing uses an 80/20 stratified train/test split with imputation and one-hot encoding in a scikit-learn Pipeline. Modeling starts with logistic regression and progresses to XGBoost and LightGBM with cross-validated hyperparameter tuning. SHAP values handle interpretation. Deliverables: GitHub repo, project report, slide deck, and a targeting-decision memo for the marketing-analytics stakeholder.